

Understand negative numbers and number lines

Step 42

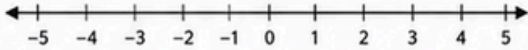
Use numbers below zero in context and reason about direction and comparison.

★ Today I am learning to understand negative numbers and number lines.

1. Number Lines

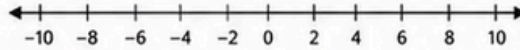
Use the number lines to answer the questions.

a) Use the number line below.



- 1) Which number is 3 steps to the left of 1? _____
- 2) Which number is 4 steps to the right of -2? _____
- 3) Plot and label the point for -4.
- 4) Plot and label the point for 2.

b) Use the number line below.



- 1) Which number is halfway between -6 and -2? _____
- 2) Which number is 5 steps to the right of -7? _____
- 3) Plot and label the point for -9.
- 4) Plot and label the point for 7.

Remember:

- Numbers to the left are smaller.
- Numbers to the right are larger.
- Zero is the point where positive and negative numbers meet.

2. Compare and Order

Compare the numbers using $<$, $>$ or $=$. Then order the sets from least to greatest.

- | | | |
|--------------------|---------------------|--|
| a) -3 _____ 2 | e) -8 _____ -8 | i) Order from least to greatest:
$-5, 2, -8, 0, 4$ _____ |
| b) -7 _____ -4 | f) -1 _____ 1 | j) Order from least to greatest:
$-10, -3, -6, 2, -1$ _____ |
| c) 0 _____ -1 | g) -10 _____ -3 | k) Order from greatest to least:
$3, -2, -7, 0, -5$ _____ |
| d) 5 _____ -2 | h) 4 _____ -5 | |

Remember:

- -7 is less than -4 .
- -3 is greater than -8 .
- Any negative number is less than any positive number.
- Zero is greater than any negative number.

3. Real-World Contexts

Read each situation and write the number that best represents it.

a) The temperature is 6 degrees below zero.



Number: _____

b) A submarine is 120 metres below sea level.



Number: _____

c) A bank account balance is \$45 overdrawn.

Account Balance
-\$45.00

Number: _____

d) An elevator goes down 3 floors from ground level.



Number: _____

e) A plane is 2,000 metres above sea level.



Number: _____

f) A hiker is 75 metres below the starting point.



Number: _____

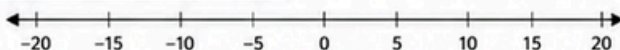
4. Problem Solving

Solve the problems. Show your working.

- | | | | |
|--|---|--|---|
| a) The temperature in the morning is -3°C . By midday, it rises 7°C . What is the temperature at midday?
Answer: _____ $^{\circ}\text{C}$ | b) Liam has \$12. He spends \$15 on lunch. What is his new balance?
Answer: \$ _____ | c) A diver is at -8 metres. He swims up 15 metres. What is his new position?
Answer: _____ metres | d) A car travels 5 km west, then 9 km east. What is its final position relative to the start?
Answer: _____ km |
|--|---|--|---|

5. Number Line Reasoning

Use the number line to help answer the questions.



- | | |
|---|--|
| a) Which is farther from 0: -12 or 8 ? _____ | d) Starting at -5 , move 9 steps to the right. What number do you land on? _____ |
| b) Which is farther from 0: -15 or 10 ? _____ | e) Starting at 7, move 14 steps to the left. What number do you land on? _____ |
| c) What number is halfway between -10 and 0 ? _____ | f) List three numbers between -6 and 3 . _____, _____, _____ |

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more help.



This was hard.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|--|--|
| ☐ Confident with negative numbers | ☐ Explains direction and movement |
| ☐ Understands number lines | ☐ Solves problems independently |
| ☐ Uses real-world contexts | ☐ Needs extra practice |
| ☐ Compares numbers accurately | |

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____

